

Protein Subcellular Localization Prediction by Combining ProtBert and BiGRU

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Abstract—The subcellular localization of proteins is very important for further understanding their functions. This paper proposes a deep learning method called PBLoc to predict the protein subcellular localization from sequence information only. In the PBLoc method, the pre-trained protein model – ProtBert, is utilized to extract the features of the protein sequence, followed by a bidirectional GRU model to predict subcellular localization. In order to capture sequence features more accurately, an attention mechanism is also used in the model. To verify the performance of our proposed PBLoc method, it is applied to DeepLoc benchmark dataset. For comparison, other 8 SOTA algorithms are also executed. The results show that PBLoc without using any evolution feature outperforms the other compared algorithms.

Index Terms—protein subcellular localization, ProtBert, BiGRU, attention mechanism

I. INTRODUCTION

Protein is the most important carrier of life activities. Abnormal localization of proteins is often associated with diseases [1] [2]. Therefore, protein localization prediction is of great significance for understanding the function and mechanism of proteins.

Recently, many deep learning methods have developed for protein subcellular localization. DeepLoc was developed by combining convolutional neural network (CNN) with long short-term memory (LSTM) [3]. Based on DeepLoc, TargetP (v2.0) further applied BiLSTM [4]. Jiang et al. put forward the MULocDeep model to realize suborganellar localization prediction [5]. Besides BiLSTM, MULocDeep used multi head self-attention mechanism. DeepMito is a specific method for mitochondrial sublocation prediction based on CNN [6]. Cong et al. presented a model based on DCNN with a self attention mechanism for multiple protein subcellular localization

prediction [7]. Most of the above mentioned deep learning methods took protein raw sequences or other simple properties as inputs. Therefore, it is important to obtain unbiased features of protein sequences without relying on evolutionary information. Elnaggar et al. proposed a pre-trained protein representation model - ProtBert [8] which is obtained via fine-tuning the original Bert [9] model with the protein sequences from UniRef100 [10] and BFD [11] databases.

This paper proposes a deep learning method combined with ProtBert and BiGRU for predicting the subcellular Localization (PBLoc) from sequence information only. In PBLoc method, ProtBert is utilized to extract the features of the protein sequence. A BiGRU [12] module is followed by ProtBert to deal with the output embeddings of ProtBert. An attention mechanism is also used to capture sequence features more accurately. The numerical results on the benchmark dataset verified the effectiveness of PBLoc.

II. METHOD

This paper proposes a ProtBert and BiGRU integrated model for the protein subcellular Localization prediction (PBLoc), which consists of four modules, namely ProtBert /embedding, BiGRU, attention, and dense/output modules, respectively. The algorithm framework is shown as Fig. 1 The four modules are described in detail below.

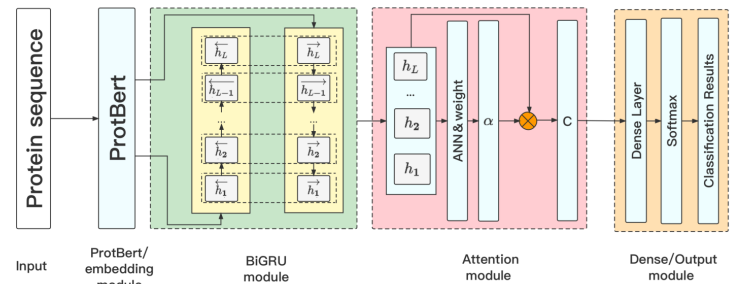


Fig. 1. The architecture of PBLoc. It consists of 4 modules, ProtBert /embedding module, BiGRU module, attention module, dense/output module.

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A. ProtBert / embedding module

In PBLoc, the pre-trained ProtBert model is utilized to extract the features of the protein sequence.

Denote the input protein sequence as X . To normalized the input length of protein, we set the input sequence as the fix length $L = 1000$. If a protein exceeded this length, the first 500 amino acids from N-terminus and the last 500 amino acids from C-terminus were preserved and combined. If a protein had a shorter sequence, it should be padded to 1000 at the end and masked the padding part for the following calculation. Then, ProtBert transfers the protein sequence into its embedding by

$$Y = \text{ProtBert}(X) \quad (1)$$

where the input $X \in \mathbb{R}^{batchsize \times L}$, the output $Y \in \mathbb{R}^{batchsize \times L \times d}$, and d is the dimension size of feature vector, respectively.

B. BiGRU module

After obtaining the embedded representation Y of the protein sequence, the BiGRU model is used for further extraction of the sequence features.

For any position l , the input data feature is $y_l \in \mathbb{R}^{d \times 1}$. Denote the forward and reverse hidden states at position l as $\vec{h}_l \in \mathbb{R}^{h \times 1}$ and $\overleftarrow{h}_l \in \mathbb{R}^{h \times 1}$, respectively, where h is the dimension size of hidden variables. Then the forward and reverse hidden variables are updated as follows:

$$\vec{h}_l = \phi(W_y^{(f)} y_l + W_h^{(f)} \vec{h}_{l-1} + b^{(f)}) \quad (2)$$

$$\overleftarrow{h}_l = \phi(W_y^{(b)} y_l + W_h^{(b)} \overleftarrow{h}_{l+1} + b^{(b)}) \quad (3)$$

where the weights $W_y^{(f)}, W_y^{(b)} \in \mathbb{R}^{h \times d}$, $W_h^{(f)}, W_h^{(b)} \in \mathbb{R}^{h \times h}$ and the bias $b^{(f)}, b^{(b)} \in \mathbb{R}^{h \times 1}$ are all model parameters that can be learned, and ϕ is the activation function of the hidden layer.

Finally,

$$h_t = \begin{bmatrix} \vec{h}_l \\ \overleftarrow{h}_l \end{bmatrix} \quad (4)$$

is obtained as the output of the BiGRU model corresponding to the input data feature y_l at position l .

C. Attention module

An attention function can be described as a mechanism that enables the module to focus on important information. Therefore, the attention mechanism is employed to find amino acids with higher attention weights.

To calculate the attention weights for each position, the output h of BiGRU module is transformed using a feed-forward neural network with tanh active function:

$$u_i = \tanh(W_w h_i + b_w) \quad (5)$$

where W_w and b_w are the learnable weights and bias parameters. Then the attention weights could be obtained by

$$\alpha_t = \frac{\sum_w \exp(u_i^T u_w)}{\sum_i \sum_w \exp(u_i^T u_w)} \quad (6)$$

At last, the original output h of BiGRU module will be summed according to the attention weights by

$$C = \sum_i \alpha_i h_i \quad (7)$$

The attention mechanism in the model is intuitively plausible. The module MULocDeep has proved that the contribution of each amino acid in localization is different, and the regions with higher attention weights are more likely to contribute to the localization. As a result, this paper holds that the neural network combined with the attention mechanism can enhance the interpretation and produce a more accurate assessment of the outcome. An ablation experiment is executed to analyze the effects of the attention module in Experiment section.

D. Dense / output module

Through all the above layers, the attention weighted feature representation C of the entire protein sequence has been obtained. In order to get the final subcellular localization classifications, a 2-layer-dense-network and a softmax layer are followed and form the final dense / output module.

The first layer of dense network utilizes sigmoid function, thereby the output of which can be expressed by

$$H = \sigma(W^{(1)} C + b^{(1)}) \quad (8)$$

where $W^{(1)}$ is the weight matrix, $b^{(1)}$ is the bias vector, and $\sigma()$ is sigmoid function. The second layer of dense network use identity function as its active function, the output of which is

$$O = W^{(2)} H + b^{(2)} \quad (9)$$

where $W^{(2)}$ and $b^{(2)}$ are the weight matrix and bias vector.

At last, the output O is normalized by softmax function to create probability vector $\hat{O}$ as follow:

$$\hat{O}_i = \frac{\exp(O_i)}{\sum_k \exp(O_k)} \quad (10)$$

The element $\hat{O}_i$ means the probability of the related protein belongs to the i th subcellular localization.

During the training process, cross-entropy loss function is used to update the parameters of the model, that is

$$J = - \sum_{i=1}^N y_i \log(q_i(x)) \quad (11)$$

where $q_i(x)$ represents the probability that the model predicts x to be classified into class i , N is the number of labels, and y_i represents whether the real label corresponding to the input x is class i .

III. EXPERIMENTAL RESULTS

A. Datasets

We train and evaluate PBLoc on the DeepLoc dataset, which is constructed by selecting protein sequences from the UniProt database according to the following criteria: eukaryotic cells; encoded in the nucleus and not fragmented; longer

than 40 amino acids. The investigation was annotated (ECO: 0000269).

The model is trained based on 5-fold cross validation method. To guarantee good generalization of the model, a stringent homology partitioning is performed and each cluster of homologous proteins is assigned to one of the five folds, ensuring that similar proteins were not mixed between the different folds.

B. Experiment Setup

Ablation tests and comparison experiments are designed to verify the effectiveness of the proposed PBLoc model. First, to analyze the performance of the BiGRU architecture in the model, it is compared to the other 2 recurrent neural network architectures (i.e. RNN, and GRU) without adding attention mechanisms. Secondly, another ablation experiment is performed to study the functions of the attention mechanism. The models with and without attention mechanism are compared by using the winner recurrent neural network architecture – BiGRU. At last, to test the performance of our proposed PBLoc model, it is compared with other 8 state-of-the-art algorithms, namely LocTree2, MuLtiLoc2, SherLoc2, YLoc, CELLO, iLoc-Euk, WoLF PSORT, and DeepLoc, respectively.

The results of 8 comparison algorithms are cited from Ref. [18]. Our proposed PBLoc model was implemented with PyTorch1.11.0 and trained on NVIDIA RTX 2080ti 10G. It utilizes the cross-entropy loss function Eq. (18), and uses the Adam SGD optimizer to train the model.

To compare with other algorithms, the accuracy metric is used to evaluate the performance of different methods. For a single class of subcellular localization, the accuracy is defined by

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} \quad (12)$$

where TP, TN, FP, FN are True Positives, True Negatives, False Positives, and False Negatives, respectively. While for multi-class situation, the accuracy is defined by

$$Accuracy = \frac{\sum_i c_{i,i}}{\sum_{j,k} c_{j,k}} \quad (13)$$

where $c_{j,k}$ is the element of confusion matrix, meaning the number of samples in the j th category but being predicted as the k th class. It is easy to see that Eq. (19) is a special case of Eq. (20).

C. Recurrent neural network layer structure analysis

To analyze the performance of the BiGRU architecture in the model, it is compared with RNN and GRU models. The results of individual location types are listed in Table I, where the bold items are the best ones among the three comparison structures. From the table, it could be found that BiGRU performs best on all subcellular localization classes on the accuracy metric. On average accuracy, the BiGRU structure significantly outperforms the RNN structure with 21.9% improvement, and slightly outperforms the GRU structure with 0.8% improvement. Fig. 2 shows the accuracy comparison

results of the whole model. We also could find that the BiGRU structure performs best, and the RNN structure is far below the other two ones, implying that the GRU structure can improve the ability of learning features from long protein sequences by refreshing the update gate frequently.

TABLE I
COMPARISON RESULTS ON DIFFERENT RECURRENT NEURAL NETWORK LAYER STRUCTURE.

Category	Accuracy		
	RNN	GRU	BiGRU
Cell membrane	0.9076	0.9553	0.9583
Cytoplasm	0.6768	0.8497	0.8605
Endoplasmic reticulum (ER)	0.6488	0.8414	0.8509
Extracellular	0.6506	0.8450	0.8533
Golgi apparatus	0.6667	0.8491	0.8587
Lysosome/Vacuole	0.6768	0.8688	0.8778
Mitochondrion	0.6327	0.8724	0.8837
Nucleus	0.6547	0.8915	0.8968
Peroxisome	0.9326	0.9720	0.9779
Plastid	0.9463	0.9917	0.9917
Average	0.7394	0.8937	0.9010

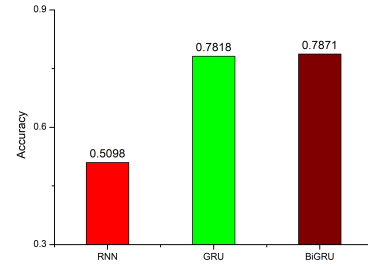


Fig. 2. Comparison of the overall performance of the three structures (RNN, GRU, BiGRU).

D. Attention mechanisms analysis

Based on the above experimental results, we chose BiGRU as the base model in our subsequent experiments. To verify the impact of the attention mechanisms on the model, a comparative experiment is executed, and the experiment results of individual location types are listed in Table II. After removing attention, the average accuracy decreases by 0.39%. Especially, the application of the attention enhances the performance of Endoplasmic reticulum (ER), Extracellular, Golgi apparatus, Lysosome/Vacuole, Mitochondrion, and Nucleus, respectively. Fig. 3 shows the comparison results of the accuracy of the whole model with and without the utilization of the attention module. It could be found that the model with attention module obtains an improvement of 2.4% compared with reducing attention module.

E. Comparison with Other Methods

Finally, to test the performance of our proposed PBLoc model, it is compared with other 8 algorithms. The comparison results are listed in Table III. For the 8 compared algorithm, their results are reported in Ref. [3]. It could be found that

TABLE II
COMPARISON RESULTS ON WITH OR WITHOUT ATTENTION MODULE

Category	Accuracy	
	PBLoc	PBLoc without Attention
Cell membrane	0.9535	0.9583
Cytoplasm	0.8557	0.8605
Endoplasmic reticulum (ER)	0.8563	0.8509
Extracellular	0.8623	0.8533
Golgi apparatus	0.8646	0.8587
Lysosome/Vacuole	0.8831	0.8778
Mitochondrion	0.8927	0.8837
Nucleus	0.9094	0.8968
Peroxisome	0.9767	0.9779
Plastid	0.9905	0.9917
Average	0.9045	0.9010

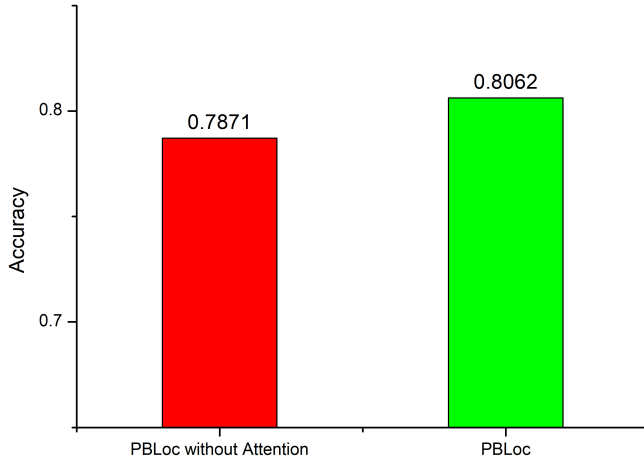


Fig. 3. Comparison of the overall performance of the difference between with Attention or not.

DeepLoc and our PBLoc are much better than the other 7 algorithms. The best one of the 7 seven algorithm is iLoc-Euk, the accuracy of which is 0.6820. The result of DeepLoc is 0.7797, with 14.3% improvement to iLoc-Euk, while the accuracy of PBLoc is 0.8062, with 3.4% improvements to DeepLoc. It should be noted that though the PBLoc performs best among all methods, it doesn't use any evolution information, and its protein sequence representation ProtBert model is offline pre-trained, requiring less computation cost.

TABLE III
COMPARISON RESULTS WITH OTHER 8 MODELS

Model	Accuracy
LocTree2	0.6120
MultiLoc2	0.5592
SherLoc2	0.5815
YLoc	0.6122
CELLO	0.5521
iLoc-Euk	0.6820
WoLF PSORT	0.5671
DeepLoc	0.7797
PBLoc	0.8062

IV. CONCLUSIONS

This paper proposes a deep learning model (PBLoc) to predict protein subcellular location by using sequence information alone. The pre-trained protein model ProtBert is utilized to enhance the protein sequence representation, followed by a bidirectional GRU module for further sequence feature extraction. The model also incorporates an attention mechanism to better capture different sequence positions' importance related to subcellular location task. The performance of our suggested PBLoc approach is tested using the DeepLoc benchmark datasets, as well as other eight SOTA algorithms in this field. The numerical results demonstrate that, in the absence of evolution features, PBLoc outperforms the other comparison algorithms.

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